

Spectral calibration and analysis

III assignment of *Astrofisica Osservativa*

Bernardo Vettori

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Abstract

The aim of this assignment consists of three parts: the wavelengths and absolute calibration of spectra, and the estimation of Jupiter's rotational period. After image calibration and spectrum extraction, the wavelength calibration is provided by a second-degree polynomial fit to the data of a reference calibration lamp. This constrains line identification in the wavelength range sampled by the lamp spectrum. Then we assume a time-independent response function of the detector and a plane parallel atmosphere to simplify the relation between the acquired signal and the density flux of the star, the atmospheric transmittance, the exposure time, and precisely the response. We estimate the response function for a set of wavelength bins using two spectra of Vega acquired at different altitudes, and a reference density flux of the star. The result is qualitative, but the density flux thus calibrated is acceptable to the reference one. Finally, the rotational velocity of Jupiter is estimated from the tilt of one of the lines: we study two spectra at the top and bottom of the line (corresponding to the ends of the equator) and take the maximum of the cross-correlation as the Doppler shift. Knowing the velocity (computed by the Doppler formula) and the radius of the planet (from the database), the computation of the rotational period is straightforward, and the result is (15 ± 2) h. The discrepancy with the reference value, about 10 h, is due to a bad data acquisition procedure. The spectral image, in fact, is not aligned with the dispersion direction. After correcting this effect, we compute the period by averaging the values estimated from five different tilted lines and obtain (10.3 ± 0.7) h, compatible with the expected value.

1 Introduction

The third assignment of *Astrofisica Osservativa* requires the implementation of a procedure to calibrate and analyze some acquired spectra. In particular, it consists of three tasks: the wavelength and flux calibration of star spectra and the estimation of Jupiter's rotational period from the Doppler shift effect.

The first part of the report revolves around the methods and procedures to prepare and manipulate data. Section 2 summarizes the features of the implemented Python package, while Section 3 explains how an acquired image is calibrated and how the spectrum is extracted from it.

After the target spectrum is obtained, we have to calibrate it in wavelength, that is, map pixel values along the dispersion direction into the corresponding wavelengths. The analysis of the spectrum of a reference lamp accomplishes the task, permitting one to identify lines in the spectrum (restricted to the wavelength range of the lamp). The procedure and results are discussed in Section 4 for α Lyræ, α Bootis and β Draconis. The second task is fulfilled by using two acquisitions of the same target at different altitudes (airmasses) during the

observation night. Section 5 reports the theoretical assumptions and the following steps to calibrate Vega's spectrum in flux. Finally, the estimation of Jupiter's rotational period from the lines tilted due to the Doppler effect is discussed in Section 6.

2 The python package

To complete the required tasks, we implemented a Python package called `spectralpy`¹. It consists of 5 modules:

- `stuff.py`
This module collects some useful methods (such as the ones to compute the mean and STD or to remove hot pixels) and classes (`FuncFit` to provide a fit, and `Spectrum` to handle spectral data).
- `display.py`
Methods to quickly plot data or `.fits` files are in this module.

¹See https://github.com/00-berni/proj_3.

- **data.py**
This module provides all the methods to open the files and extract the data. It also collects paths of files and directories.
- **calcorr.py**
This is the main module of the package. It is used for the computation of Master Bias, Master Dark, and Master Flat and for the wavelength calibration.

The package documentation contains a detailed description of each function and module.

In addition to that, a script per each task was implemented: `targ_spec.py`, `abs_cal.py`, and `jup_period.py`, respectively.

3 Data

The Teams channel provided a set of observations along different years². The ones used in this report were acquired at the observatory of Punta Falcone (LI): star targets on 27th March 2017 using the spectroscope Alpy 600 and Jupiter on 22nd April 2018 using the spectroscope Lhires III. The data in 2017 also include five acquisitions of Bias, Dark, and Flat.

3.1 Image calibration

Before the spectrum is extracted, two steps are required:

1. Compute the science frame from the row image acquired by the CCD, using the calibration frames.
2. Correct the slit inclination by rotating the image.

Science frame First, we compute master bias $\mathcal{B}$, master dark $\mathcal{D}$, and master flat $\bar{\mathcal{F}}$ [2, 4]:

$$\begin{aligned}\mathcal{B} &\equiv \frac{1}{5} \sum_{i=0}^5 \mathcal{B}_i \\ \mathcal{D} &\equiv \frac{1}{5} \sum_{i=0}^5 \mathcal{D}_i - \mathcal{B} \\ \mathcal{F} &\equiv \frac{1}{5} \sum_{i=0}^5 \mathcal{F}_i - \mathcal{B} - \mathcal{D} \\ \bar{\mathcal{F}} &\equiv \langle \mathcal{F} \rangle_{(x,y)}\end{aligned}$$

where the i index represents the i th acquisition and $\bar{\mathcal{F}}$ is the flat normalized by the average over the pixels inside the slit. The Dark acquisition has the highest exposure time, so before subtracting it from any frame, it has to be divided by its exposure time and multiplied by the

frame's one³. Following [2–4], the science frame (SF) is computed as:

$$\text{SF} = \frac{\mathcal{I} - \mathcal{B} - \mathcal{D}}{\bar{\mathcal{F}}}$$

where $\mathcal{I}$ represents either the Light Frame of the target or its Master Light, that is the average of all Light Frames of the target. The user can choose between the two.

Inclination correction Before extracting the spectrum, the inclination of the slit has to be corrected. The method `angle_correction()` selects a column every 10 pixels, for each one finds the position of the brightest pixel and sets the relative HWHM along the cross-dispersion direction as the uncertainty. A linear fit to the points then provides the estimation of the angle of inclination.

To refine this estimation, the routine is run again after the rotation of the image, but the script fits a Gaussian to the brightness profile of each column instead of just looking for the positions of the maximum values. Then, the means of the Gaussian profiles are used to compute the tilt angle by a linear fit. Two examples of image rotation are reported in Figures 18 and 22 for α Bootis and Jupiter, respectively.

We do not use this procedure for calibration lamps: the image is just rotated by the same angle as estimated for the corresponding target. The assumption is that the lamp was acquired immediately after the target, so they are tilted the same way.

3.2 Spectrum extraction

The following analysis requires a one-dimensional spectrum. The method `spectrum_average()` provides that: first, as the previous routine, it collects the positions of the brightest pixels and the corresponding HWHM values and then computes a band centered on the average of the estimated heights and broad the mean HWHM. Figure 2 (top) shows an example. The extracted spectrum is obtained by averaging the rows⁴ within this band along the cross-dispersion direction, and the corresponding STD is taken as the uncertainty.

This method is not suitable for the calibration lamp, so we average only rows within an interval centered at the mid-height of the brightest line, as shown in Figure 2 (bottom).

To better estimate the band within which the spectrum is extracted, a mask was first applied to the images, removing the background light outside the slit.

³The linearity of the dark current with exposure time [2–4] allows for this procedure.

⁴Each row is normalized along the dispersion direction by default. However, the method allows the user to choose whether to perform this step or not.

²I grouped all of them in the folder `data_files` and listed in the file `objs_per_night.json`

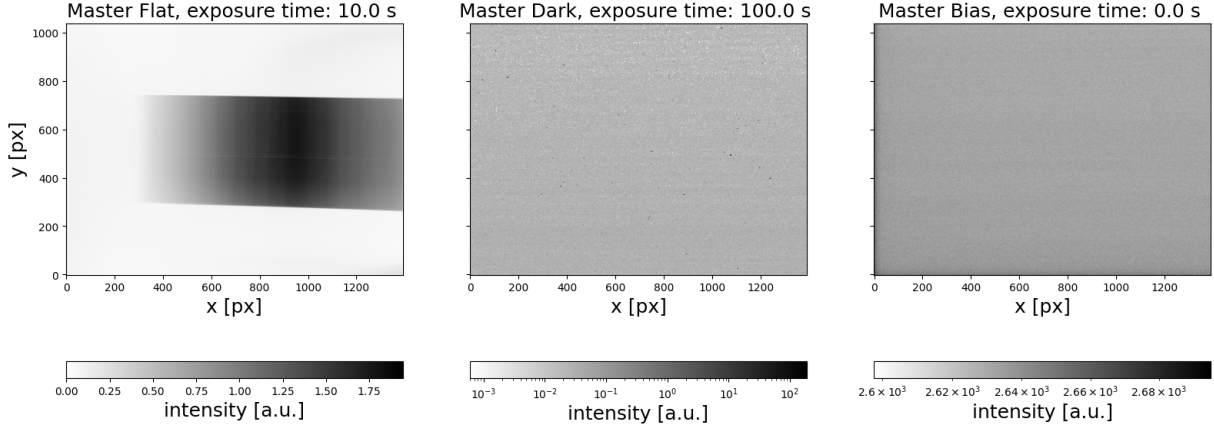


Figure 1: Calibration frames used to calibrate the spectral image.

4 Wavelength calibration

The first of the three tasks of the assignment requires wavelength calibration of the spectra. The calibration lamp was acquired for this purpose. The reference lines of the used NeAr lamp and the extracted spectrum of the Vega lamp are shown in Figure 19. We chose only 24 lines, which are the ones reported as single lines, and the corresponding pixel positions are collected in the file `data_file/Vega01/calibration_lines.txt`. To estimate the calibration function $F_\lambda(x_{px})$, that is, the function to map pixels to angstroms, the method `lines_calibration()` provides a fit with a second-degree polynomial. The estimated parameters are listed in Table 1 and the result is shown in Figure 3. The dispersion of the residuals is taken as the uncertainty in wavelengths estimated from $F_\lambda(x_{px})$.

Fit parameters of $F_\lambda(x_{px})$	
a	$(-2.18 \pm 0.07) \cdot 10^{-4} \text{ \AA/px}^2$
b	$(3.81 \pm 0.01) \text{ \AA/px}$
c	$(2810 \pm 5) \text{ \AA}$

Table 1: Results of the calibration fit of $F_\lambda(x_{px}) = ax_{px}^2 + bx_{px} + c$.

A second-degree polynomial was chosen because using a linear fit produced structured residuals while using a higher-degree polynomial gave parameters with too high uncertainties (about 30% or more). The spectrum of Vega thus calibrated is shown in Figure 4.

4.1 Lines identification

The procedures discussed so far are all collected in the method `calibration()`, which takes the observation night, the number of the acquisition, and the target name as input values and returns the calibrated spectra of both the target and the corresponding lamp.

The identification lines procedure is provided by the script `targ_spec.py`, instead. The script calls the method `calibration()` but restricts the wavelength range to the same one as the calibration lamp. The reason is that this range is well sampled and it guarantees to be consistent. The uncertainty associated with the wavelengths is not low enough to do a proper identification, so only the main lines of the spectral type are reported. Using the NIST Atomic Spectra Database Lines⁵ it is possible, for example, to identify $H\alpha$ and $H\beta$ in the spectrum of Vega, as shown in Figure 5.

4.1.1 Other targets

In addition to α Lyræ, the spectra of α Bootis and β Draconis were calibrated and they are shown in Figures 6 and 7, respectively.

To avoid passing through the estimation of the calibration function for each new target, the method `calibration()` permits the comparison of the lamp used for Vega with the new one. The two lamps could be shifted one relative to the other due to a change in the tilt angle of the slit during the observation night. Hence, the method computes the cross-correlation between the lamps, stores the lag for which the correlation is the maximum, and shifts the new target's data by that value. The results are reported in Figure 20.

We follow [6] to check the identified lines.

4.1.2 Telluric contamination

Each spectrum displays the same broad lines in the range from 6600 $\text{\AA}$ to 7700 $\text{\AA}$ as shown in Figure 21. These are due to the absorption of Earth's atmosphere molecules [1, 6], like O_2 and H_2O , and are not related to the stars.

⁵https://physics.nist.gov/PhysRefData/ASD/lines_form.html

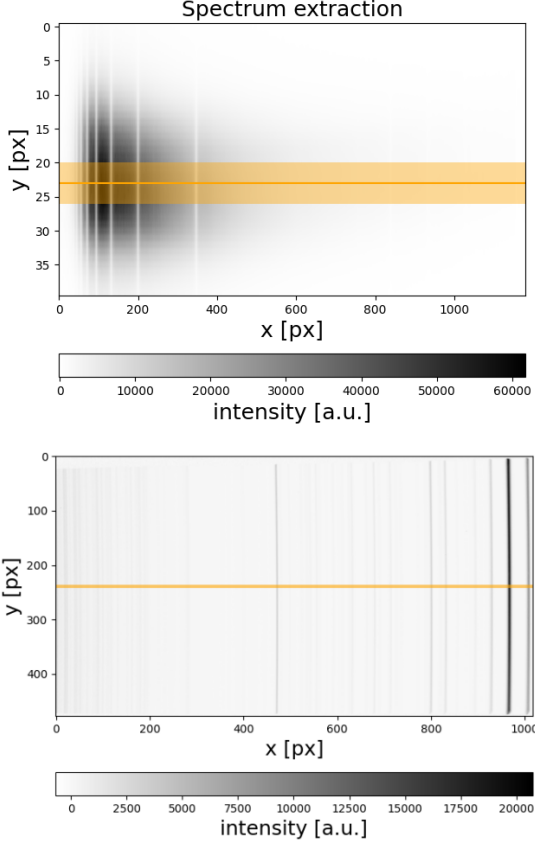


Figure 2: Spectral images of α Lyræ (top) and the corresponding lamp (bottom). The spectrum is obtained by averaging the pixels within the yellow band along the cross-dispersion direction.

5 Absolute calibration

5.1 Theoretical background

Calibrating the spectra in flux requires the estimation of the detector's response function. The measured spectrum $\mathcal{N}_\lambda$ can be written as [1, 3–5]:

$$\mathcal{N}_\lambda(X, t) = \mathcal{T}_\lambda(X) \mathcal{S}_\lambda \mathcal{R}_\lambda(t) t$$

where X is the airmass, t is the exposure time, $\mathcal{T}_\lambda$ is the atmospheric transmittance, $\mathcal{S}_\lambda$ is the spectral flux density of the target and $\mathcal{R}_\lambda$ is the response function. Assuming $\mathcal{R}_\lambda(t) = \mathcal{R}_\lambda$ and plane parallel atmosphere $\mathcal{T}_\lambda = \exp(-\tau_\lambda X)$ [3–5], we get:

$$\mathcal{N}_\lambda(X, t) = e^{-\tau_\lambda X} \underbrace{\mathcal{S}_\lambda \mathcal{R}_\lambda}_{\Sigma_\lambda} t \quad (1)$$

where τ_λ is the optical depth and Σ_λ is the spectrum acquired outside the atmosphere. Then from (1) knowing Σ_λ and $\mathcal{S}_\lambda$ for a certain source the response function is just:

$$\mathcal{R}_\lambda = \frac{\Sigma_\lambda}{\mathcal{S}_\lambda} \quad (2)$$

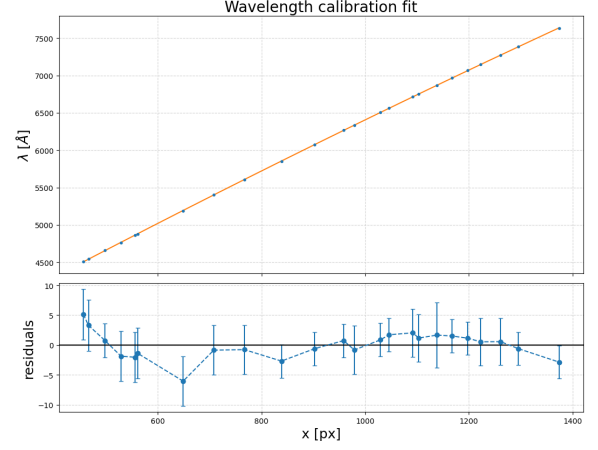


Figure 3: Second-degree polynomial fit (top) and the relative residuals (bottom) for the estimation of the calibration function.

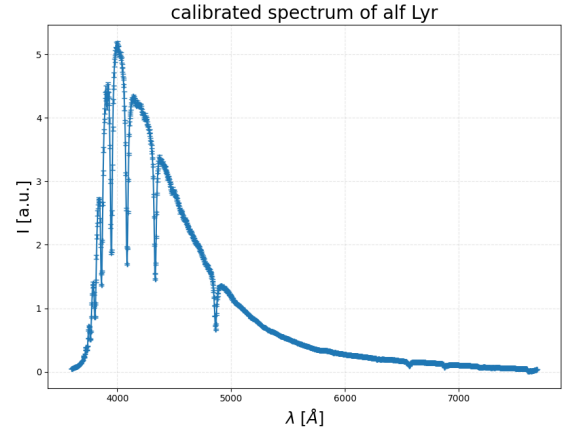


Figure 4: Spectrum of Vega calibrated in wavelength.

5.2 Setup

We use two acquisitions of Vega at different altitudes taken during the same night. Data are listed in Table 2. The script implemented to accomplish this task is `abs_cal.py`.

UT (28/03/2017)	alt_i [deg]	X_i
00:30:29	34.63 ± 0.01	$1.7595 \pm 4 \cdot 10^{-4}$
01:24:04	43.86 ± 0.01	$1.4432 \pm 3 \cdot 10^{-4}$

Table 2: Selected data for the estimation of the response function. The uncertainty in the altitude is set a priori. The airmass is defined as [3–5] $X_i = \csc(alt_i)$

The first step is to extract the spectra and calibrate them in wavelengths, therefore the script calls the method `calibration()`, but it does not require normalization in brightness as before, because we are interested in the acquired counts. Then, to be able to compare dif-

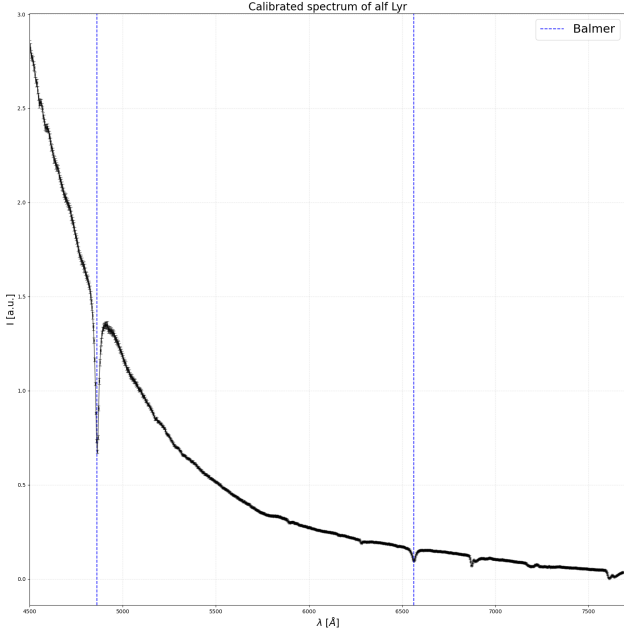


Figure 5: The blue lines are H_α (6562.79 Å) and H_β (4861.350 Å) detected in Vega spectrum.

ferent spectra, we perform a data binning by dividing the wavelength range from 4500 Å to 6500 Å into 50 Å intervals and averaging the counts within each bin. Before this step, however, the script removes the strongest lines (Balmer series) from both spectra, interpolating⁶ the values in the gaps, to avoid spurious results. The data normalized by the exposure time are shown in Figure 8.

5.3 Zero-airmass spectrum

To estimate Σ_λ requires to compute $\mathcal{T}_\lambda$. From the assumption of the plane parallel atmosphere and no strong stratification in the thermal properties, the optical depth can be modeled by an exponential law [4]:

$$\tau_\lambda = f_\tau(\lambda) = e^{\mu(\lambda - \lambda_0)} + \tau_0 \quad (3)$$

Using the two acquisitions, the value of the optical depth in each bin λ_k can be computed by:

$$\tau_k = \ln \left(\frac{\mathcal{N}_{k,2} t_1}{\mathcal{N}_{k,1} t_2} \right) \frac{1}{X_1 - X_2}$$

Table 3 and Figure 9 report the results of the best-fit of (3) to data (λ_k, τ_k) .

At this point, we can compute Σ_λ by averaging $\mathcal{N}_{\lambda,i} \exp(-f_\tau(\lambda)X_i)/t$ over the two observations. The result is shown in Figure 10.

⁶We use a cubic spline interpolation, provided by the class `CubicSpline` of the package `scipy.interpolate`.

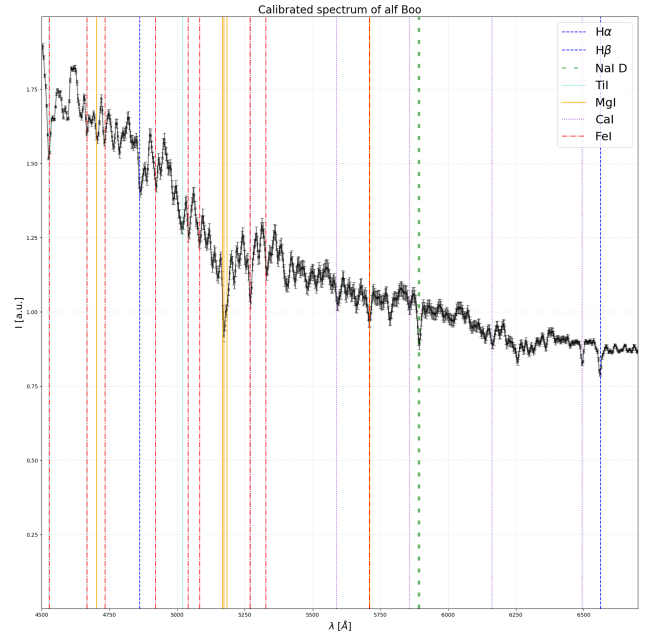


Figure 6: Lines detection for α Bootis.

Fit parameters	
μ	$(-9.0 \pm 0.7) \cdot 10^{-4} \text{ Å}^{-1}$
λ_0	$(4.26 \pm 0.02) \cdot 10^3 \text{ Å}$
τ_0	0.24 ± 0.03

Table 3: Results of the fit of (3).

5.4 Response function estimation

The last step is to estimate the response by (2), so the flux density of Vega is necessary. The used standard was taken from <https://ftp.eso.org/pub/stecf/standards/hststan/>, but we perform the same manipulations on the data as on the spectra, that is Balmer's lines removal and data binning. The standard spectrum $\mathcal{S}_k^{std}$ and the response function $\mathcal{R}_k$ thus estimated are shown in Figures 11 and 12, respectively.

5.5 Standard vs Calibrated

Finally, we qualitatively compare the chosen standard spectrum with the one thus calibrated $\bar{\mathcal{S}}_k$. The latter is computed by:

$$\bar{\mathcal{S}}_k = \frac{1}{2\mathcal{R}_k} \sum_{i=1}^2 \frac{\mathcal{N}_{k,i}}{t_i} \exp \left(f_\tau(\lambda_k) X_i \right)$$

From Figure 13 we note that the discrepancy between the two is less than 0.31%, far from H_β .

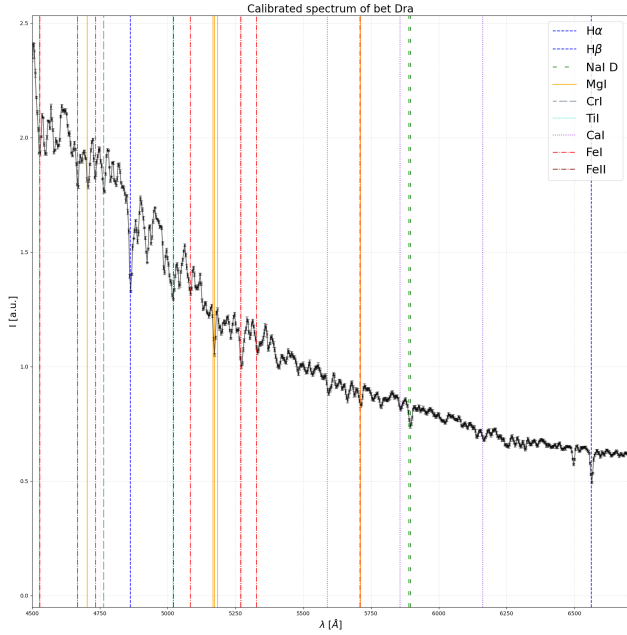


Figure 7: Lines detection for β Draconis.

6 Jupiter's period

The estimation of Jupiter's rotational period requires (a) to calibrate the image, (b) to calibrate in wavelengths the spectrum, and (c) to compute the Doppler effect. These three are provided by the script `jup.period.py`.

6.1 Image calibration

As with the previous targets, Jupiter's spectral image must be rotated. The procedure is the same as before except for the first step: after the identification of maxima along the cross-dispersion direction, the script discards points outside a chosen strip in the inner part of the spectral image. This prevents spurious results because the previous method applied in broad absorption lines gives points farther from the central spine of the spectral image.

6.2 Wavelength calibration

The calibration lamp spectrum samples a wavelength range around the H α (the broadest line in Jupiter's spectrum). Calibration lines are saturated and there are, moreover, only three. The former problem can be easily managed by taking the position of the middle of the saturated part as the line peak position. The latter, instead, obliges a linear fit for wavelength calibration to avoid over-fitting. The results are reported in Table 4 and Figure 23. Reference wavelength values are taken from the manual⁷.

⁷[data_files/calibration/sques-relco-sc480-calibration-lines-3.0.pdf](#).

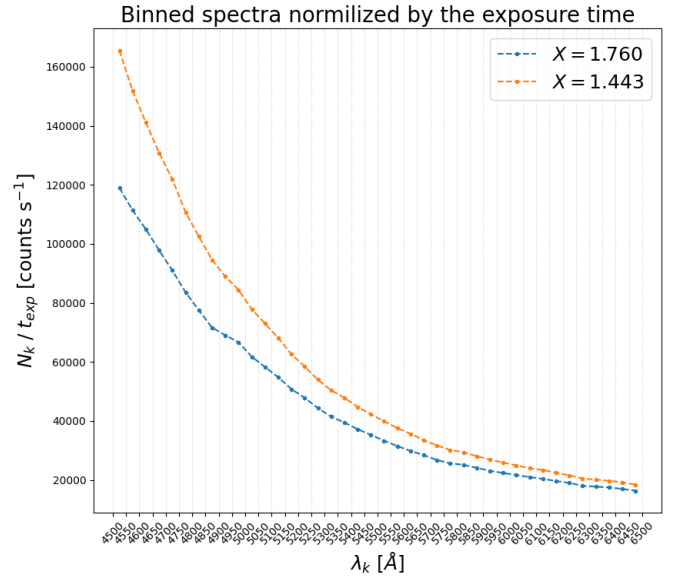


Figure 8: The two acquisitions of Vega after data binning and exposure time normalization.

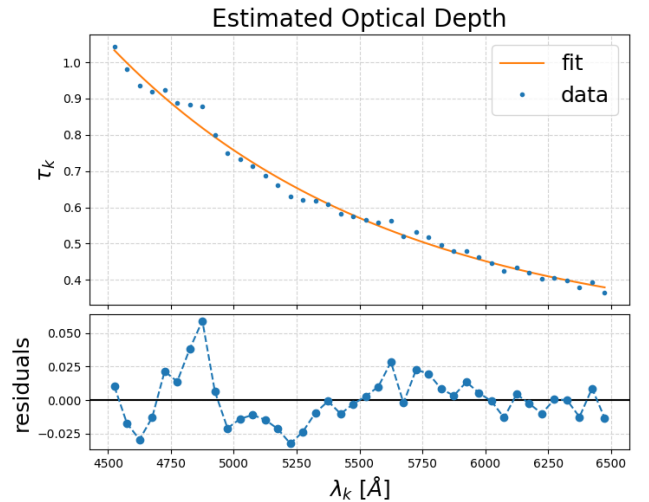


Figure 9: Results of optical depth fit.

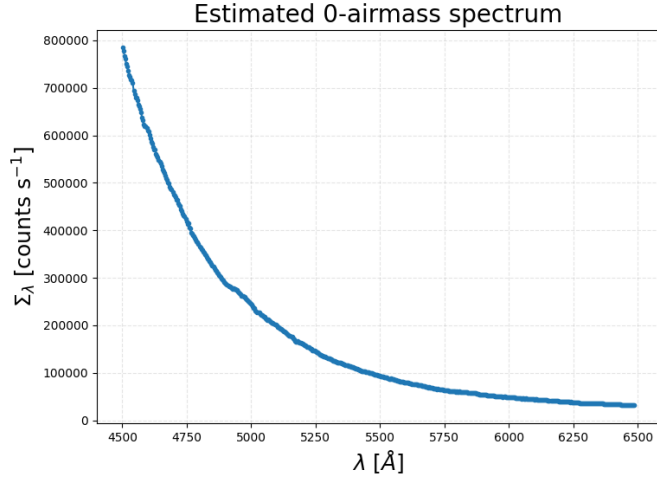


Figure 10: Average zero-airmass spectrum of Vega.

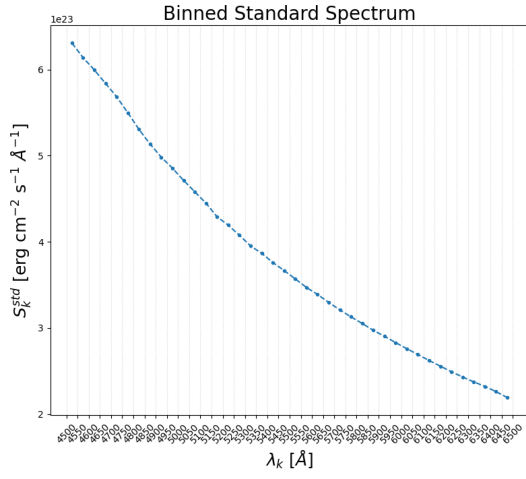


Figure 11: Standard spectrum of α Lyræ after Balmer's lines removal and data binning.

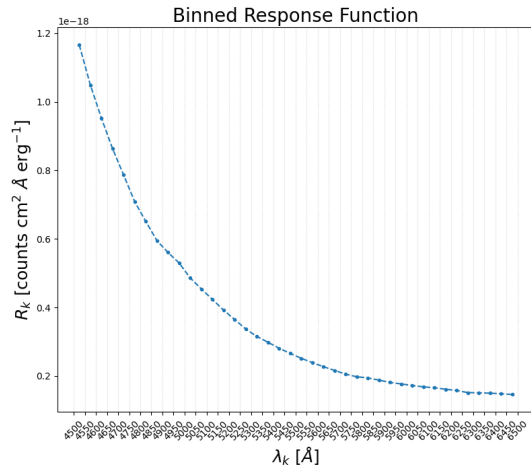


Figure 12: Estimated response function for each bin.

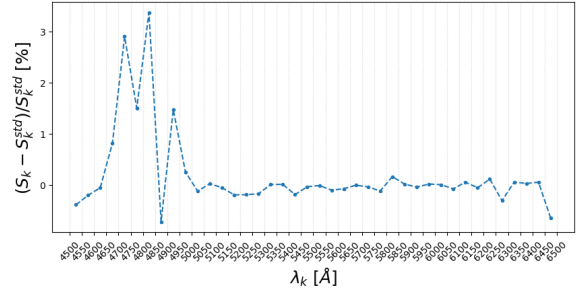
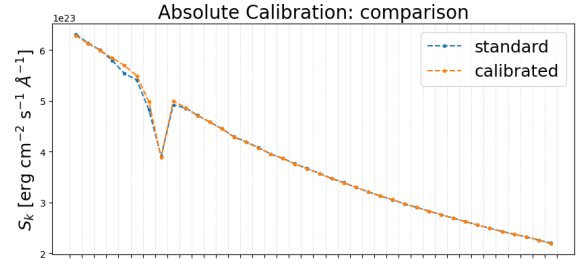
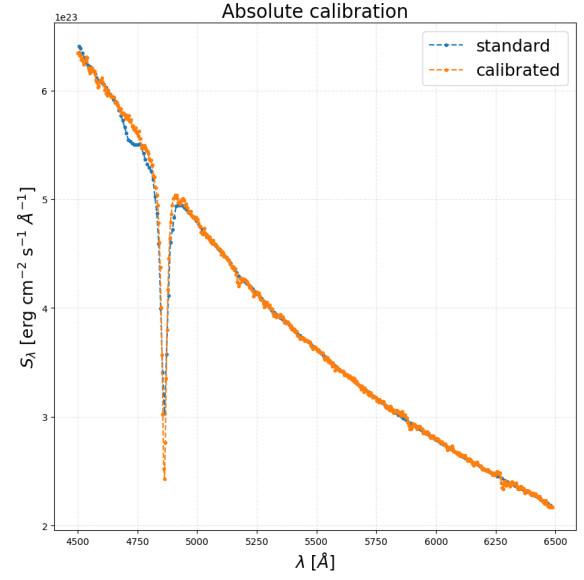


Figure 13: Comparison between the standard spectrum and the one obtained by the calibration before (top) and after (bottom) data binning.

Fit parameters of $F_\lambda(x_{px})$	
m	$(7.993 \pm 0.004) \cdot 10^{-2} \text{ \AA/px}$
q	$(6495 \pm 3) \text{ \AA}$

Table 4: Results of the linear fit of $F_\lambda(x_{px}) = m x_{px} + q$ for the wavelength calibration of Jupiter.

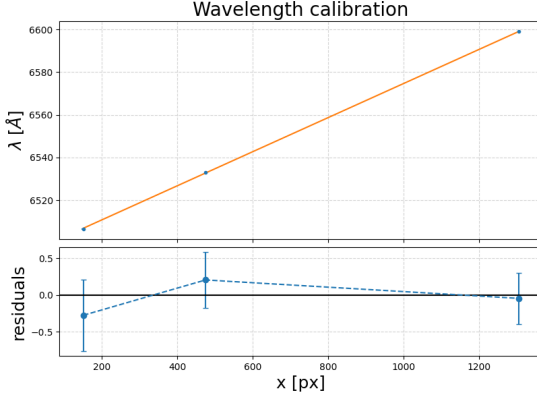


Figure 14: Results of the wavelength calibration for Jupiter.

6.3 Period estimation

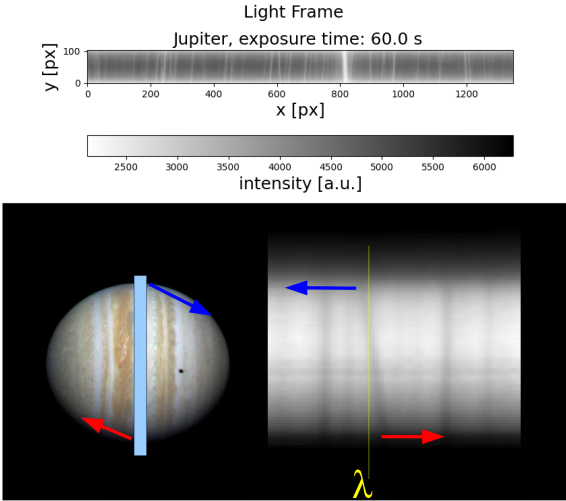


Figure 15: Top: acquired spectral image of Jupiter. Lines are tilted due to the Doppler shift effect. Bottom: Scheme to explain this effect.

During the acquisition of the target, the slit was aligned with the equator of the planet, as shown in Figure 15. This ensures to estimate the value of rotational velocity from the tilt of the lines due to the Doppler effect:

$$v_{rot} = \frac{c}{2} \frac{\delta\lambda}{\lambda_0}$$

where c is the speed of light in vacuum, λ_0 is the central wavelength of a line, and $\delta\lambda$ is the Doppler shift

in wavelength. The factor $1/2$ takes into account that light is shifted twice⁸. In reality, we do not compute $\delta\lambda$ but $\Delta\lambda = 2\delta\lambda$, that is, the difference between the central wavelength values at the top and bottom of the spectral image for a certain line. Assuming to know Jupiter's radius R_J , then the period is:

$$T_J = \frac{2\pi}{\Omega_J} = 8\pi \frac{R_J}{c} \frac{\lambda_0}{\Delta\lambda} \quad (4)$$

Used referenced values are reported in Table 5.

Name	Value
Radius R_J	71492 km
Ang. Diameter θ_J	44.29871 arcsec
Sid. Rot. Period T_J	9 ^h 55 ^m 29.711 ^s

Table 5: Reference values of Jupiter from [Horizons Web Application](#). Radius is measured at the equator and angular diameter is on the date and at the observation site.

Regarding the line to be studied, $H\alpha$ is not a good candidate because of its thickness, so we choose the most tilted of the other lines instead. The pixel p_0 corresponding to λ_0 of the line is the position of the minimum value of the cumulative profile, that is, the sum pixel-to-pixel along the cross-dispersion direction, shown in Figure 16. To choose properly the two spectra to be compared the script `jup_period.py` estimates first the central axis along the dispersion direction. The procedure consists of fitting a Gaussian to the profile of a column every 10 pixels, computing the position of its mean, storing the value, and averaging all heights thus estimated. The selected spectra are those at a distance of the angular radius⁹ of the planet from the central axis previous estimated. The shift Δp of one row relative to the other is the one that maximizes the cross-correlation between the two spectra and it is the Doppler shift in pixels.

All results are reported in Table 6. Thanks to the calibration in wavelength it is possible to pass from pixel to the corresponding wavelength value: $\lambda_0 = m p_0 + q$ and $\Delta\lambda = m \Delta p$. Then (4) becomes:

$$T_J = 8\pi \frac{R_J}{c} \frac{p_0 + q/m}{\Delta p}$$

The estimated period is $(15 \pm 2) \text{ h}$.

⁸Solar light is blue(red)-shifted first when it "hits" Jupiter's atmosphere and then when it is reflected.

⁹This value is obtained from the angular diameter of Jupiter θ_J on the date and at the site of the observation (see Table 5). Known the focal length f of the telescope and the physical size of pixels s_{px} , the angular radius in pixels is just $\rho_{px} = (\theta_J/2)f/s_{px}$ with θ_J in radians. For this acquisition $f = 2880 \text{ mm}$ and $s_{px} = 6.44922 \mu\text{m}$, so $\rho_{px} \simeq 48 \text{ px}$.

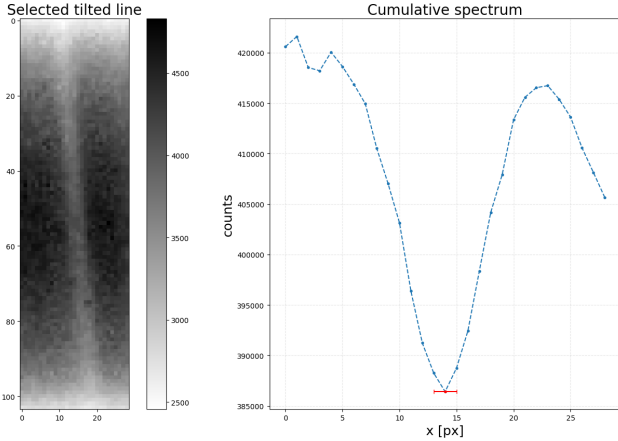


Figure 16: Left: the most tilted line used for the period estimation. Right: the cumulative spectrum of the line, that is, the sum along the cross-dispersion direction. The red point is the central pixel p_0 .

Name	Value [px]
p_0	711 ± 1
Δp	9 ± 1

Table 6: The uncertainties are set to 1 px nominally.

6.4 Misalignment correction

The estimated period is incompatible with the reference and the discrepancy is about 3σ (see Table 5). The reason is that the adopted procedure is invalidated by a mistake in acquiring data. We expect the spectrum to display straight lines due to terrestrial atmospheric absorption, but as shown in Figure 15 they are tilted¹⁰ instead. We assume this effect to be purely geometrical due to a bad acquisition procedure. So, after rotating the spectrum again to correct it, we select five different lines and estimate the period for each one. The method is similar to the previous one, except for $\Delta\lambda$, which is computed this time by the difference between the minimum positions at the top and bottom of each line. The selected lines are shown in Figure 17 and the values thus obtained are listed in Table 7. The rotational period of Jupiter is the average of them:

$$\bar{T}_J = (10.3 \pm 0.7) \text{ h}$$

which is compatible with the reference.

We do not know how the acquisition procedure was performed. One possible explanation could be that to align the slit with Jupiter’s equator the angle between the spectrograph and the slit was not maintained fixed, causing misalignment of the image with the dispersion direction.

¹⁰They are tilted in the opposite direction of Jupiter’s lines, i.e. $H\alpha$.

i	$p_{0,i}$ [px]	$T_{J,i}$ [h]
1	632	11 ± 1
2	919	9.8 ± 0.7
3	1220	8.6 ± 0.6
4	1232.5	9.1 ± 0.6
5	1278	12 ± 1

Table 7: Selected lines. For each one, the central pixel and the estimated period are reported. $p_{0,3}$ and $p_{0,4}$ are computed by averaging the minimum positions at the top and bottom of the lines, instead of taking the central pixel of the cumulative spectrum. The reason is that these lines are closed enough to generate a single broad line in the summed spectrum.

Finally, there is also another detail that we ignored during the procedure, which is the discrepancy of about 1.8σ between the value of $H\alpha$ and its estimation from the spectrum, which is $\lambda_{H\alpha}^{est} = (6562.3 \pm 0.3) \text{ \AA}$. This may be related to the bulk motion of the planet relative to the Earth. From the Doppler shift effect we obtain $v_{bulk} = (-11 \pm 6) \text{ km/s}$, which is in accordance with the value -9.2583743 km/s from [Horizons Web Application](#), but the resolution is not enough to trust this result. Maybe three calibration lines are not enough to sample the wavelength range. However, taking into account this effect does not change the result.

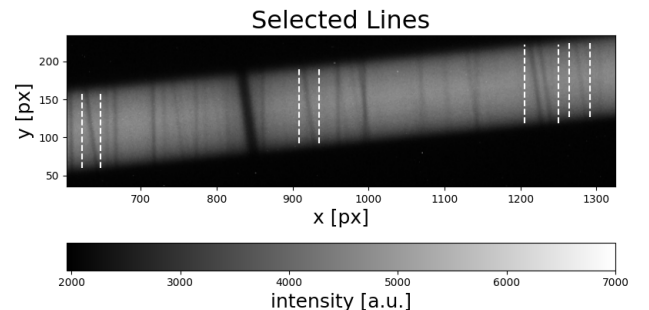


Figure 17: Lines selected to compute the rotational period.

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- [6] Richard Walker. *Spectral Atlas for Amateur Astronomers: A Guide to the Spectra of Astronomical Objects and Terrestrial Light Sources*. Cambridge University Press, 2017.

Image Rotation

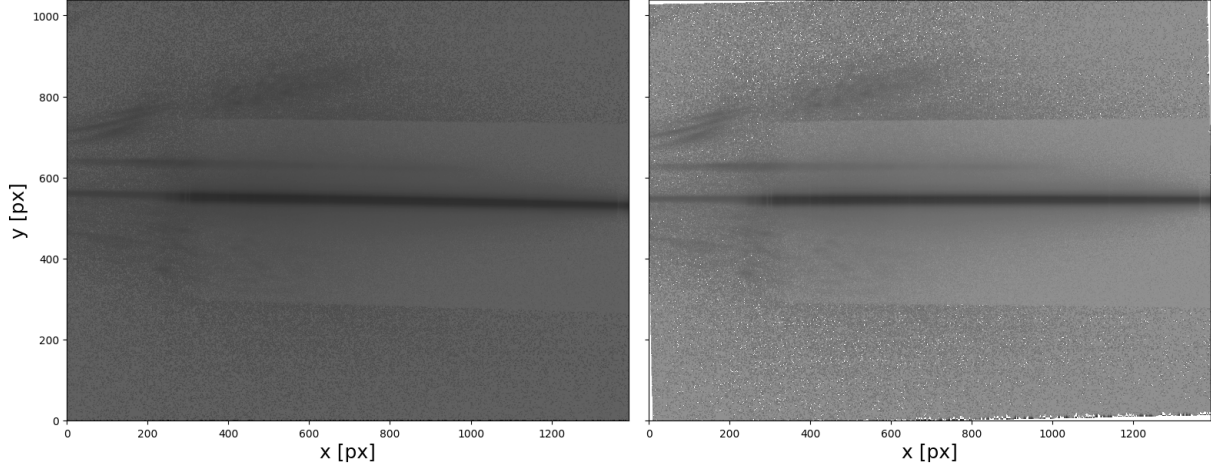


Figure 18: Example of image rotation for α Bootis

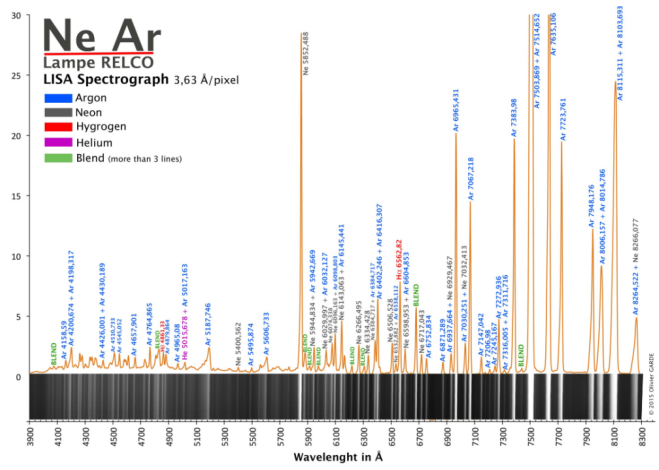
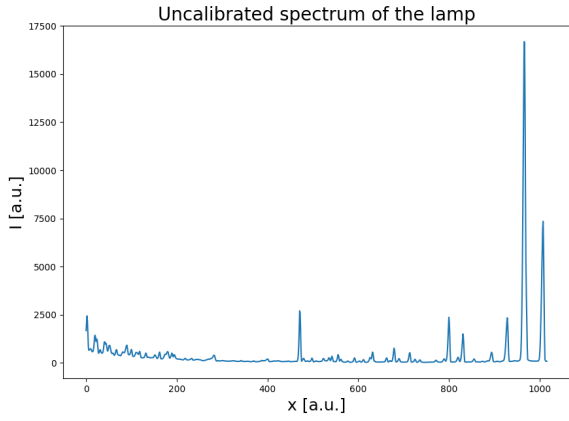


Figure 19: Spectrum of calibration lamp for Vega (left) and the reference (right). Single lines only were chosen.

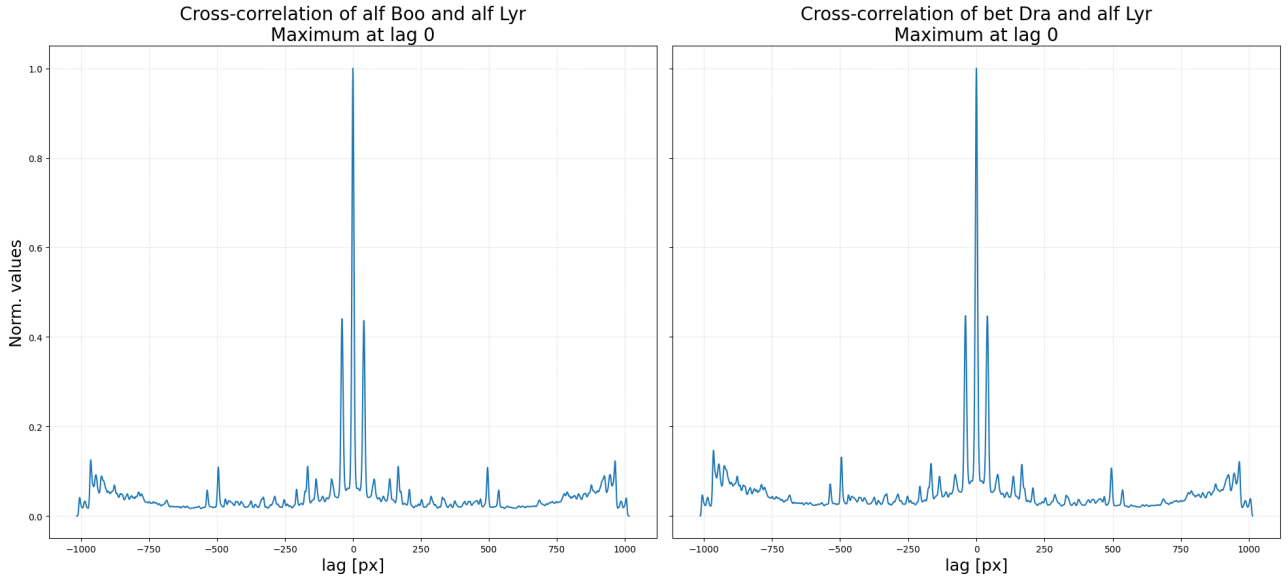


Figure 20: Cross-correlation between the lamps of different targets. Both cases do not require to shift of the spectrum relative to the one of the Vega lamp, because the correlation is the maximum when the data overlap.

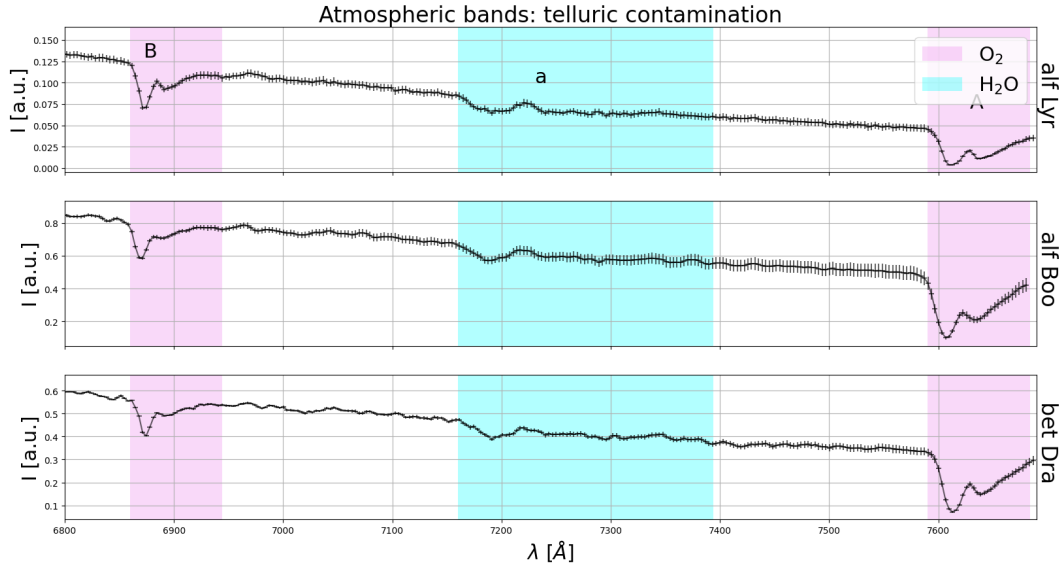


Figure 21: Absorption lines due to Earth's atmosphere in the spectra. We can identify the Fraunhofer A and B bands of O_2 [1, 6].

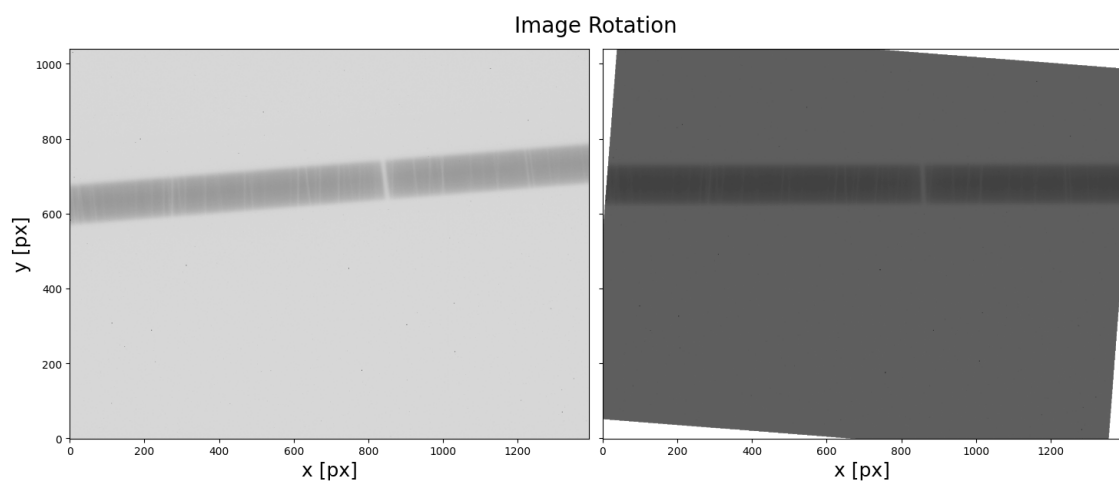


Figure 22: Jupiter image rotation.

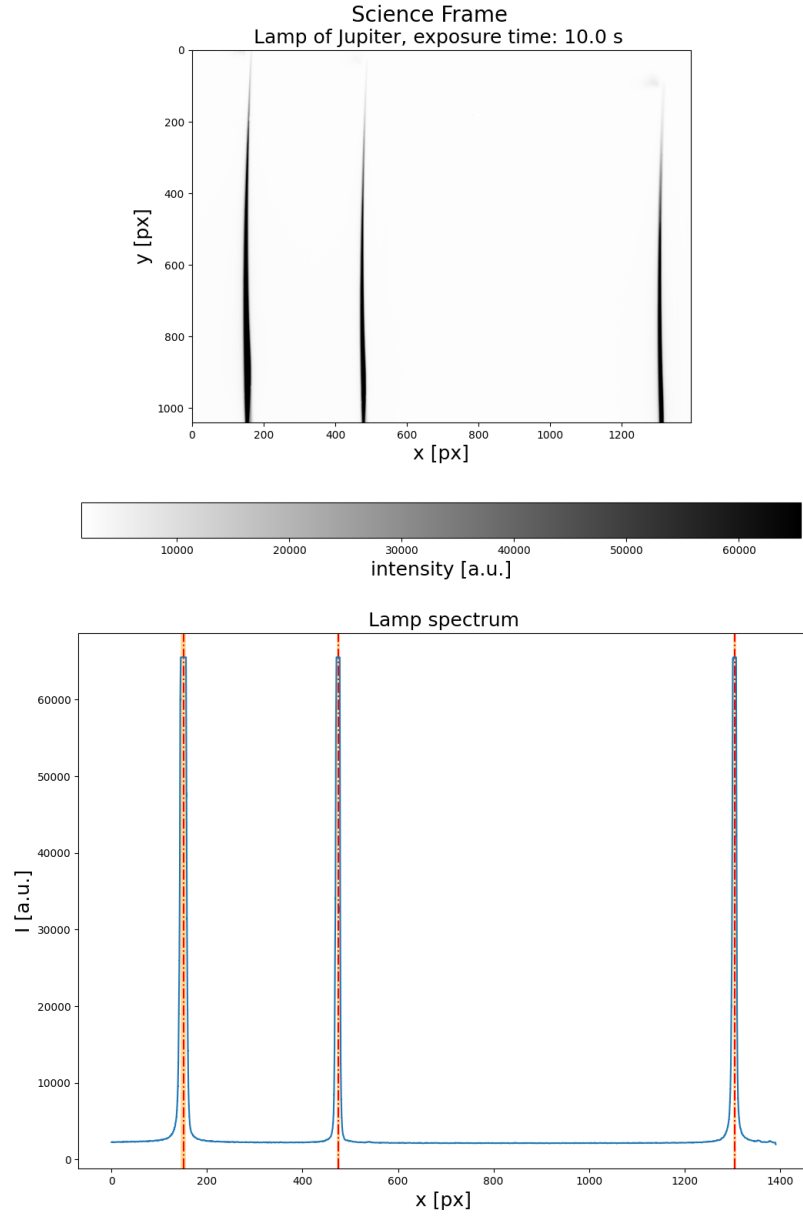


Figure 23: Spectral image (top) and extracted spectrum (bottom) of the Jupiter's lamp. Red lines are the central values and the yellow bands are the uncertainties.